

1.

a.


 $Q_{matrix} =$

$$\begin{matrix}
 & 0 & 1 & 2 & 3 & 4 & \dots \\
 \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{bmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots \\ \mu & -(\mu+\lambda) & \lambda & 0 & 0 & \dots \\ 0 & 2\mu & -(2\mu+\lambda) & \lambda & 0 & \dots \\ 0 & 0 & 3\mu & -(3\mu+\lambda) & \lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}
 \end{matrix}$$

b. Writing the rate balance equations; we have;

$$\pi_{n-1} \lambda = n\mu \pi_n \quad \forall n \geq 1$$

$$\pi_n = \left(\frac{\lambda}{n\mu}\right) \pi_{n-1} \quad \forall n \geq 1$$

So;

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!} \pi_0$$

So;

$$\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!} \pi_0 = 1$$

 $\Rightarrow$

$$\pi_0 = \frac{1}{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n}$$

let $\lambda/\mu = \rho$. From Taylor's expansion;

$$\sum_{n=0}^{\infty} \frac{1}{n!} \rho^n = e^{\rho}$$

So,
(A solⁿ exists and is
therefore the stationary
distribution)

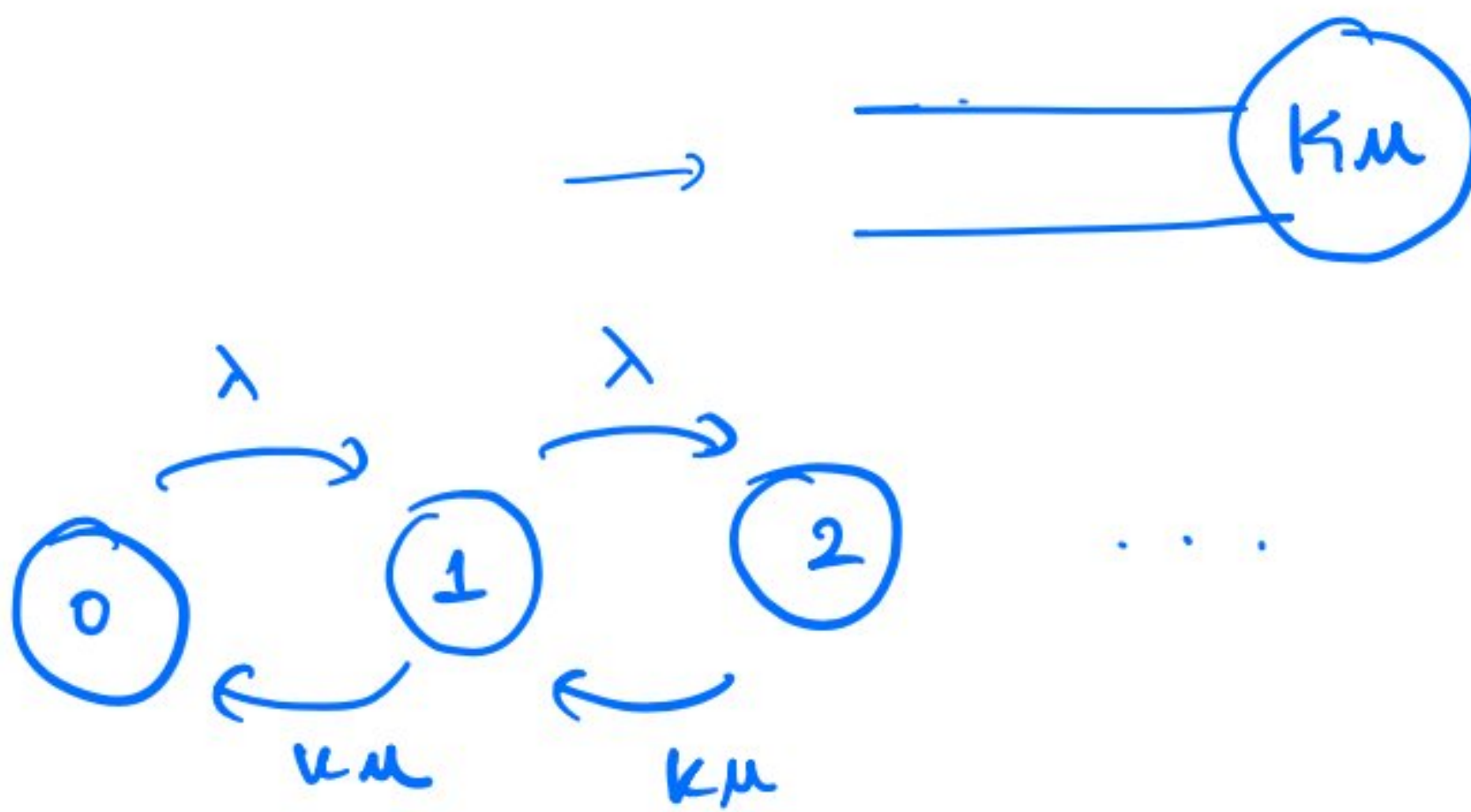
$$\Rightarrow \pi_0 = e^{-\rho}$$

$$\Rightarrow \pi_n = \frac{\rho^n}{n!} e^{-\rho}$$

$\therefore$ The stationary distribution follows a poisson distribution with parameter γ_μ .

2.

(a) $M/M/1/\infty$



Rate balance to obtain stationary dist:

$$\pi_0 \lambda = \pi_1 k\mu \Rightarrow \pi_1 = \pi_0 (\lambda/k\mu)$$

$$\pi_1 \lambda = \pi_2 k\mu \Rightarrow \pi_2 = \pi_1 (\lambda/k\mu) = \pi_0 (\lambda/k\mu)^2$$

$$\vdots$$

$$\pi_{n-1} \lambda = \pi_n k\mu \Rightarrow \pi_n = \pi_{n-1} (\lambda/k\mu) = \pi_0 (\lambda/k\mu)^n$$

$$\therefore \pi_n = \pi_0 \left(\frac{\lambda}{k\mu} \right)^n$$

$$\sum_{i=0}^{\infty} \pi_i = 1 \Rightarrow \sum_{i=0}^{\infty} \pi_0 \left(\frac{\lambda}{k\mu} \right)^i = 1$$

$$\Rightarrow \pi_0 \sum_{i=0}^{\infty} \left(\frac{\lambda}{k\mu} \right)^i = 1$$

$$\Rightarrow \pi_0 \left(\frac{1}{1 - \frac{\lambda}{k\mu}} \right) = 1$$

$$\Rightarrow \pi_0 = 1 - \frac{\lambda}{k\mu} = 1 - \rho$$

$$E(N) = \sum_{n=0}^{\infty} n \pi_n = \sum_{n=0}^{\infty} n \pi_0 \left(\frac{\lambda}{k\mu} \right)^n$$

$$= \pi_0 \sum_{n=0}^{\infty} n \rho^n \quad (\rho = \frac{\lambda}{k\mu})$$

$$\text{Let } S = \sum_{n=0}^{\infty} n \rho^n = \rho + 2\rho^2 + 3\rho^3 + \dots$$

$$\rho S = \rho^2 + 2\rho^3 + \dots$$

$$(1 - \rho)S = \rho + \rho^2 + \rho^3 + \dots = \frac{\rho}{1 - \rho}$$

$$S = \frac{\rho}{(1 - \rho)^2}$$

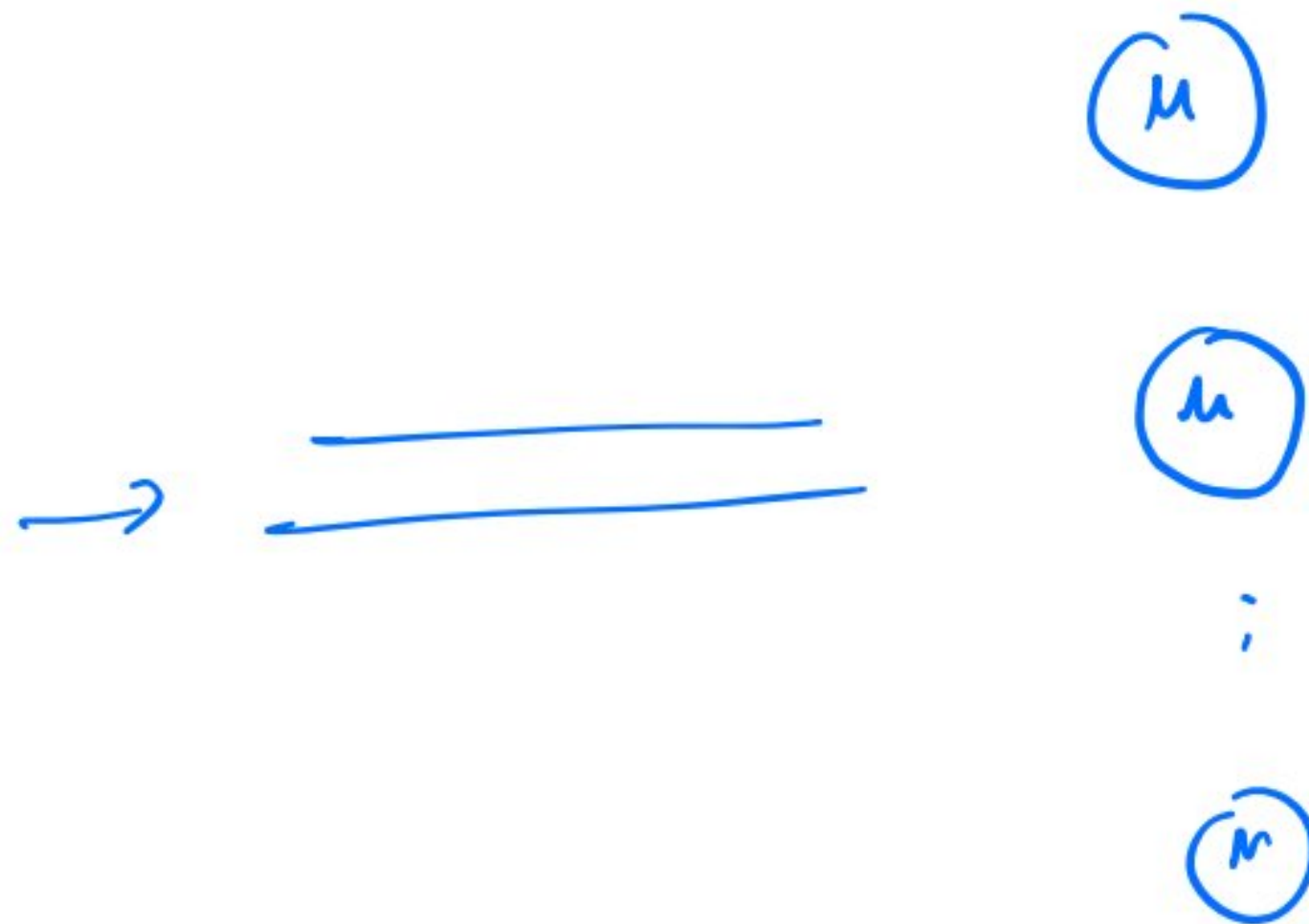
So,

$$E(N) = \frac{\pi_0 \cdot \rho}{(1 - \rho)^2} = \frac{(1 - \rho) \rho}{(1 - \rho)^2}$$

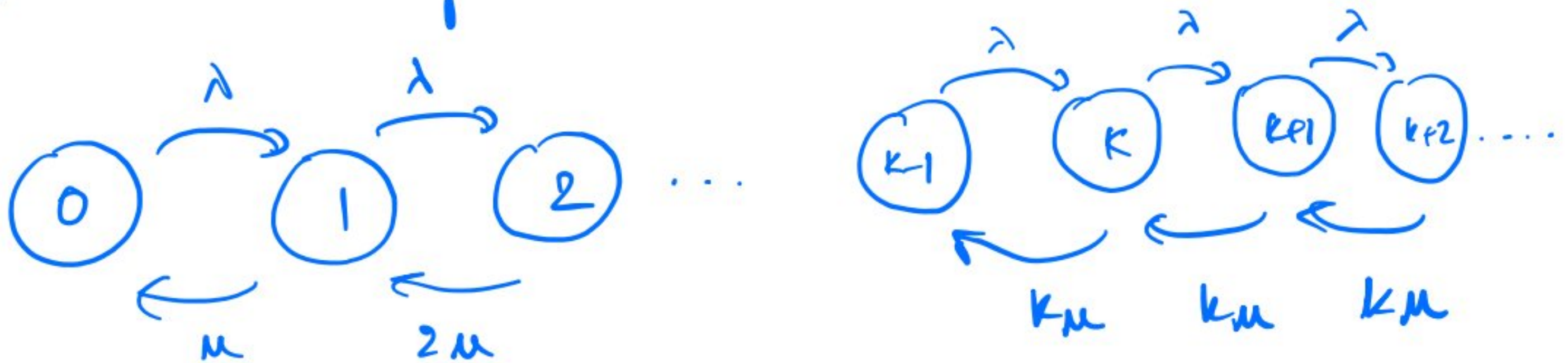
$$E(N) = \frac{\rho}{1 - \rho} \quad (\text{as } \pi_0 = 1 - \rho)$$

$$E(N) = \left(\frac{\lambda}{k\mu - \lambda} \right) = \frac{\rho}{1 - \rho}$$

b. $M/M/K/\infty$



State transition diagram:



Rate balance:

$$\pi_0 \lambda = \pi_1 \mu$$

$$\pi_1 \lambda = \pi_2 (2\mu)$$

⋮

$$\pi_{k-1} \lambda = \pi_k (k\mu)$$

$$\pi_k \lambda = \pi_{k+1} (k\mu)$$

$$\pi_{k+1} \lambda = \pi_{k+2} (k\mu)$$

⋮

$$\Rightarrow \pi_1 = \pi_0 \left(\frac{\lambda}{\mu} \right)$$

$$\Rightarrow \pi_2 = \pi_1 \left(\frac{\lambda}{2\mu} \right) = \pi_0 \left(\frac{\lambda}{\mu} \right)^2 \frac{1}{2!}$$

$$\Rightarrow \pi_3 = \pi_2 \left(\frac{\lambda}{3\mu} \right) = \pi_0 \left(\frac{\lambda}{\mu} \right)^3 \frac{1}{3!}$$

⋮

$$\Rightarrow \pi_k = \pi_{k-1} \left(\frac{\lambda}{k\mu} \right) = \pi_0 \left(\frac{\lambda}{\mu} \right)^k \frac{1}{k!}$$

$$\Rightarrow \pi_{k+1} = \pi_k \left(\frac{\lambda}{k\mu} \right) = \pi_0 \left(\frac{\lambda}{\mu} \right)^{k+1} \frac{1}{(k+1)!}$$

⋮

$$\pi_n = \pi_{n-1} \left(\frac{\lambda}{k\mu} \right) = \pi_0 \left(\frac{\lambda}{\mu} \right)^n \frac{1}{k!} \frac{1}{k^{n-k}}$$

So;

$$\pi_i = \begin{cases} \pi_0 \left(\frac{\lambda}{\mu}\right)^i \frac{1}{i!} & i \leq k \\ \pi_0 \left(\frac{\lambda}{\mu}\right)^i \frac{1}{k!} \frac{1}{k^{i-k}} & i > k \end{cases}$$

$$\sum_{i=0}^{\infty} \pi_i = 1$$

$$\Rightarrow \sum_{i=0}^k \pi_i + \sum_{i=k+1}^{\infty} \pi_i = 1$$

$$\Rightarrow \pi_0 \left(\sum_{i=0}^k \left(\frac{\lambda}{\mu}\right)^i \frac{1}{i!} + \sum_{i=k+1}^{\infty} \left(\frac{\lambda}{\mu}\right)^i \frac{1}{k!} \frac{1}{k^{i-k}} \right) = 1$$

$\xrightarrow{\frac{\rho^{k+1}}{1-\rho}}$

$$\pi_0 \left(\sum_{i=0}^k \frac{(k\rho)^i}{i!} + \frac{k^k}{k!} \sum_{i=k+1}^{\infty} \left(\frac{\lambda}{\mu k}\right)^i \right) = 1$$

where
 $\left(\rho = \frac{\lambda}{k\mu}\right)$

$$\pi_0 = \left(\sum_{i=0}^k \frac{(k\rho)^i}{i!} + \frac{(k\rho)^k}{k!} \cdot \frac{\rho}{1-\rho} \right)^{-1}$$

$$E(N) = \sum_{i=0}^{\infty} n \cdot \pi_n$$

} Simplifying till here is sufficient.

(c) By the splitting property of Poisson process, arrival to each queue individually will be a Poisson process with rate $\frac{\lambda}{K}$. Each Queue operates indep. as M/M/1.

Proof: Net arrival is according to Poisson process.

$$P(N(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

Let $N_i(t)$ rep. number of jobs in i^{th} queue @ t

→ Define p_i ($= \frac{1}{K}$ for all i) to be prob. that a randomly arrived job chooses queue i .

$$P(N_i(t) = k \mid N(t) = n) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } 0 \leq k \leq n$$

$$P(N_i(t) = k) = \sum_{n=0}^{\infty} P(N_i(t) = k \mid N(t) = n) \cdot P(N(t) = n)$$

$$= \sum_{n=0}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} \cdot \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

↓ simplify.

$$= e^{-(\lambda p)t} \frac{(\lambda p t)^k}{k!}$$

[Some justification of taking $\frac{\lambda}{K}$ is needed. Mentioning the splitting property is sufficient].

∴ This is equivalent to $K M/M/1$ with $\lambda/K \neq \mu$.

$$E(N_i) = \frac{\frac{(\frac{\lambda}{K})}{\mu}}{1 - \frac{(\frac{\lambda}{K})}{\mu}} = \frac{\rho}{1-\rho} \text{ for all } i.$$

$$E(N) = K E(N_i) \\ = \frac{K\rho}{1-\rho}.$$

Now;

$$a) E(N_A) = \frac{\rho}{1-\rho}$$

$$\text{So, } E(N_A) < E(N_c)$$

$$c) E(N_c) = \frac{K\rho}{1-\rho}$$

Comparison with b is tricky and all valid arguments will be considered.

$$E(N_a) = \frac{\left(\sum_{i=0}^{\infty} i p^i \right)}{\left(\sum_{i=0}^{\infty} p^i \right)}$$

$$\boxed{\rho = \frac{\lambda}{\mu K}}$$

$$E(N_b) = \frac{\sum_{i=1}^K i p^i \frac{K^i}{i!} + \sum_{i=K+1}^{\infty} i p^i \frac{K^K}{K!}}{\sum_{i=1}^K p^i \frac{K^i}{i!} + \sum_{i=K+1}^{\infty} p^i \frac{K^K}{K!}}$$

↳ The exact relation 'seems' to depend on K .

Explanations in Performance Modelling and
Design of CS book by Mor-Harchol

Pg 264 (There is a derivatⁿ and good intuition as why this has to be true there)

Case 1: $\beta \approx 0$

$$\begin{aligned} E(T_A) &< E(T_B) \\ E(N_A) &< E(N_B) \end{aligned} \quad \begin{array}{l} \text{f so} \\ \text{(Little's Law)} \end{array}$$

Case 2: $\beta \approx 1$

$$E(N_A) \approx E(N_B)$$

3) Def 1: The counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process having rate $\lambda > 0$, if

• $N(0) = 0$

• $N(t)$ has independent increments

• $P(N(t+s) - N(s) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$

Since it doesn't depend on s ,
Stationary increments

Def 2: The counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process having rate $\lambda > 0$, if

• $N(0) = 0$

• $N(t)$ has stationary & independent increments

• $P(N(h) = 1) = \lambda h + o(h)$

• $P(N(h) \geq 2) = o(h)$

Def 3: The counting process $\{N(t), t \geq 0\}$ is a Poisson process with rate $\lambda > 0$ if

• $N(0) = 0$

• $N(t)$ has stationary & independent increments

• Interarrival times (T_i) are iid and exponentially distributed with parameter λ

→ Refer to Sheldon Ross's theorem 2.1.1 for proof of Def 1 $\Leftrightarrow$ Def 2. and your notes :)

→ Proof for Def 1 $\Leftrightarrow$ Def 3 will be given here. From this we can say Def 2 $\Leftrightarrow$ Def 3

→ Def 1 ⇒ Def 3:

define, $T_i = \inf \{ t : N(t) = i \}$ - ~~inf~~ $\inf \{ t : N(t) = i \}$

$$T_1 = \inf \{ t : N(t) = 1 \}$$

notice $\{T_1 > t\} = \{N(t) = 0\}$

$$\Rightarrow P(\{T_1 > t\}) = P(\{N(t) = 0\})$$

$$= e^{-\lambda t} \left(\because P(N(t) = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!} \right)$$

from Def 1

$$\Rightarrow P(T_1 \leq t) = 1 - e^{-\lambda t}$$

↳ CDF of ~~exponential~~ $\exp(\lambda)$

$$\therefore T_1 \sim \exp(\lambda)$$

Now, $P(T_n > t | T_{n-1} = s)$

$$= P(N(s+t) - N(s) = 0) = e^{-\lambda t}$$

$$P(T_n > t | T_{n-1} = s) = e^{-\lambda t}$$

~~for independence,~~

~~$$P(T_1 > s, T_{n+1} > t) = P(N(s) = n, N(s+t) = n)$$~~

~~$$P(T_1 > s, T_2 > t) = P(N(s) = 0, N(s+t) = 1)$$~~

~~$$= P(N(s) = 0) \cdot P(N(s+t) - N(s) = 1)$$~~

~~$$= P(N(s) = 0) \cdot P(N(t) = 1)$$~~

$$P(T_n > t) = \int_s e^{-\lambda t} f_{T_{n-1}}(s) ds = e^{-\lambda t} \int_s f_{T_{n-1}}(s) ds$$

$$= e^{-\lambda t}$$

$$\Rightarrow P(T_n > t) = e^{-\lambda t} \Rightarrow T_n \sim \text{Exp}(\lambda)$$



Def 3 $\Rightarrow$ Def 1:

~~Def~~ define $S_n = \sum_{i=1}^n T_i \sim \text{Erlang}(n, \lambda)$

$$\Rightarrow N(t) = \max\{n : S_n \leq t\}$$

$$\{N(t) = n\} = \{S_n \leq t < S_{n+1}\}$$

$$\Rightarrow P(N(t) = n) = P(S_n \leq t < S_{n+1})$$

$$= P(S_n \leq t) - P(S_{n+1} \leq t)$$

$$\text{CDF of } S_n, F_{S_n}(t) = 1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}$$

$$\Rightarrow P(N(t) = n) = \left(1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}\right) - \left(1 - e^{-\lambda t} \sum_{k=0}^n \frac{(\lambda t)^k}{k!}\right)$$

$$\boxed{P(N(t) = n) = \frac{(\lambda t)^n}{n!} \cdot e^{-\lambda t}}$$

$$A) \sum_i \pi_i = 1 \quad \& \quad \pi_i q_{ij} = \pi_j q_{ji} \quad (\text{assuming } n \text{ states})$$

$$(a) \vec{\pi} Q = [\pi_1 \pi_2 \pi_3 \dots \pi_n] \begin{bmatrix} q_{11} & q_{12} & \dots & q_{1n} \\ \vdots & \vdots & & \vdots \\ q_{ni} & \dots & \dots & q_{nn} \end{bmatrix}$$

$$= [\Lambda_1, \Lambda_2, \Lambda_3, \dots, \Lambda_n]$$

$$\text{where } \Lambda_i = \sum_j \pi_j q_{ji}$$

$$= \sum_j \pi_j q_{ji} = \pi_i \sum_j q_{ji} \rightarrow 0$$

$$= 0$$

$$\therefore \Lambda_i = 0 \quad \forall i$$

$$\Rightarrow \vec{\pi} Q = \vec{0}$$

$\therefore$ Given $\vec{\pi}$ is a stationary distribution.

(b) A state i in a DTMC is Transient if

$$F_{ii} < 1 \Leftrightarrow \sum_n P_{ii}^n < \infty$$

A state i in a DTMC is Recurrent if

$$F_{ii} = 1 \Leftrightarrow \sum_n P_{ii}^n = \infty$$

• Positive recurrent:

$$\mu_{ii} < \infty$$

• Null recurrent:

$$\mu_{ii} = \infty$$